1. Markov Chains

Definition 1.1. The **period** of state $x \in S$ is $d(x) := \gcd(J_x)$, where $J_x := \{n \geq 1 : P^n(x,x) > 0\}$. A Markov chain $\{X_n\}$ is **aperiodic** if $d(x) = 1$ for all $x \in S$.
Note that if $P(x,x) > 0$, then $d(x) = 1$.
Proposition 1.2. *J_m is closed under addition, contains $kd(x)$ for all sufficiently large k, and period is a class property.*
Remark 1.3.

- If $\{X_n\}$ is irreducible, then to check aperiodicity, it suffices to check $d(x) = 1$ for *any* $x \in S$.
- If $\{X_n\}$ is aperiodic, then $P^n(x,x) > 0$ for any large enough n (depending on x).
- For any Markov chain $\{X_n\}$, we can construct the following "lazy" Markov chain $\{\tilde{X}_n\}$ with transition matrix

$$\tilde{P}(x,y) := \begin{cases} \frac{1}{2}P(x,y), & x \neq y \\ \frac{1}{2} + \frac{1}{2}P(x,x), & x = y \end{cases},$$

  which is always aperiodic since $\tilde{P}(x,x) > 0$ for each $x \in S$.

$\diamond$

Definition 1.4. States $x, y \in S$ **communicate** ($x \leftrightarrow y$) if there exists $m, n \geq 0$ such that $P^m(x,y) > 0$, $P^n(y,x) > 0$. The equivalence classes induced by communication are called **communication classes**.
Definition 1.5. Let $\pi : S \to [0,1]$ be such that $\sum_{x \in S} \pi_x = 1$. We say that π is a **stationary/invariant distribution** for the Markov chain $\{X_n\}$ if $\pi_y = \sum_{x \in S} \pi_x P(x,y)$ for each $y \in S$.
Proposition 1.6. *Let T_x be the number of times $\{X_n\}$ visits x. We have $\mathbb{E}[T_x | X_0 = x] = \sum_{n \geq 0} P^n(x,x)$.*

1.1. Discrete Time, Finite State Space.

Definition 1.7. A communication class $C \subset S$ is said to be **recurrent** if $P(x,y) = 0$ for each $x \in C$ and each $y \notin C$. Otherwise, C is said to be **transient**.
We say $\{X_n\}$ is **irreducible** if there is only one communication class. (In that case it must be recurrent.)
Proposition 1.8. *Assume S is finite. If C is a recurrent communication class, then*

$$\mathbb{P}\left[\exists \text{ infinitely many } n \text{ such that } X_n = y | X_0 = x\right] = 1, \quad \forall x, y \in C.$$

If C is a transient communication class, then with probability 1, *$\{X_n\}$ eventually leaves C and never returns.*
Remark 1.9. In the two cases above we have respectively:

$$\mathbb{P}\left[X \text{ hits } y \text{ before time } j \,|X_0 = x\right] \geq 1 - (1-q)^{j/n} = 1 - e^{-cj},$$

$$\mathbb{P}\left[\{X_n\} \text{ exits } C \text{ before time } j \,|X_0 = x\right] \geq 1 - e^{-cj}.$$

$\diamond$

Definition 1.10. $\tau \in \mathbb{N}_0 \cup \{+\infty\}$ is a **stopping time** if for each $n \in \mathbb{N}$, the $\{\tau = n\}$ is determined by $X_0, \ldots, X_n$.
Proposition 1.11. *Suppose $X_0 = x \in S$ is visited by $\{X_n\}$ infinitely often with probability* 1, *let τ_k be the kth time n such that $X_n = x$, and set $\tau_0 = 0$. Then the increments $\left\{(X_{\tau_k}, \ldots, X_{\tau_{k+1}})\right\}_{k \in \mathbb{N}_0} \in \cup_{j=1}^{\infty} S^j$ are iid. Thus, if y is such that $x \leftrightarrow y$, then the number of times we visit x before y has a geometric distribution.*
Theorem 1.12. *Suppose S is finite and $\{X_n\}$ is irreducible and aperiodic. Then there exists a unique stationary distribution π given by:*
Assume $X_0 = z \in S$ and set $T = \min\{n \geq 1 : X_n = z\}$. For $x \in S$, let $\tilde{\pi}(x) := \mathbb{E}\left[\#\{n \in \{0, \ldots, T-1\} : X_n = x\}\right]$. Let $\pi_x := \tilde{\pi}_x / \mathbb{E}[T]$. Moreover,

$$\lim_{n \to \infty} P^n(x,y) = \pi_y, \quad \forall x, y.$$

1.2. Discrete Time, Countable State Space.

Definition 1.13. $\{X_n\}$ is **irreducible** if for each $x, y \in S$, there exists $n \geq 0$ such that $P^n(x,y) > 0$. A state $x \in S$ is **recurrent** if $\mathbb{P}\left[\exists \text{ infinitely many } n : X_n = x | X_0 = x\right] = 1$, and **transient** otherwise.
Note that irreducibility no longer imply recurrence.
Proposition 1.14. *If $\{X_n\}$ is irreducible, then either all states are recurrent or all states are transient. In particular, it makes sense to say that $\{X_n\}$ is **recurrent** or **transient**.*
Definition 1.15. Let $\{X_n\}$ be irreducible and recurrent. We say $\{X_n\}$ is **null recurrent** if $\lim_{n \to \infty} P^n(x,y) = 0$ for each $x, y \in S$. We say $\{X_n\}$ is **positive recurrent** otherwise.
Proposition 1.16. *State x is recurrent if and only if $\sum_{n=0}^{\infty} P^n(x,x) = \infty$. Moreover, if x is transient, then with probability* 1, *the Markov chain $\{X_n\}$ visits x only finitely many times. Thus $\mathbb{P}(\exists$ infinitely many $n : X_n = x | X_0 = x)$ is either* 0 *or* 1.
Proposition 1.17. *Assume $\{X_n\}$ is irreducible. Then the following are equivalent:*

(i) *$\{X_n\}$ is positive recurrent.*
(ii) *$\{X_n\}$ has a stationary distribution.*
(iii) *$\limsup_{n \to \infty} P^n(x,y) > 0$ for all (equivalently, some) $x, y \in S$.*
(iv) *If $T_x := \min\{n \geq 1 : X_n = x\}$. Then $\mathbb{E}[T_x | X_0 = x] < \infty$ for all $x \in S$.*

Furthermore, if $\{X_n\}$ is aperiodic and positive recurrent, then the stationary distribution is unique and

$$\pi_y = \lim_{n \to \infty} P^n(x,y) = \frac{1}{\mathbb{E}\left[T_y \middle| X_0 = y\right]}, \quad \forall x, y \in S.$$

Finally, $\{X_n\}$ is null recurrent if and only if $\mathbb{P}[T_x < \infty | X_0 = x] = 1$ and $\mathbb{E}[T_x | X_0 = x] = \infty$.

1.3. Continuous Time, Finite State Space.

Definition 1.18. $\{X_t\}$ is **irreducible** if $P_t(x,y) > 0$ for some (equivalently, all) $t > 0$ and all $x, y \in S$.
Definition 1.19. We say $\pi : S \to [0,1]$ with $\sum \pi_x = 1$ is a **stationary distribution** of the continuous time Markov chain $\{X_t\}$ if whenever $X_0 \sim \pi$, we have $X_t \sim \pi$ for all $t \geq 0$.
Definition 1.20. The **time t transition probabilities** of the continuous time Markov chain $\{X_t\}$ are defined as

$$P_t(x,y) := \mathbb{P}\left[X_t = y | X_0 = x\right], \quad x, y \in S, \quad t \geq 0.$$

The **infinitesimal generator** A is

$$A_{x,y} := \begin{cases} \alpha(x,y), & x \neq y \\ -\alpha(x), & x = y \end{cases} \in \mathbb{R}^{N \times N}.$$

The **associated discrete time Markov chain** of $\{X_t\}$ has transition probabilities

$$P(x,y) = \begin{cases} \frac{\alpha(x,y)}{\alpha(x)}, & x \neq y, \\ 0, & x = y. \end{cases}$$

From the distribution of the discrete time Markov chain, we cannot recover the rates $\alpha(x,y)$ since the inter-jump times $\tau_n - \tau_{n-1}$ are lost.
Proposition 1.21. *P_t and A satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}t} P_t = P_t A, \quad P_0 = I, \implies P_t = e^{tA} = \sum_{n=0}^{\infty} \frac{(tA)^n}{n!}.$$

Proposition 1.22. *π is a stationary distribution of $\{X_t\}$ if and only if $\pi A = 0$ and $\sum_{x \in S} \pi_x = 1$.*
Proposition 1.23. *If $\{X_t\}$ is irreducible, then there exists a unique stationary distribution π for $\{X_t\}$, and*

$$\lim_{t \to \infty} P_t(x,y) = \pi_y, \quad \forall x, y \in S.$$

Discrete Time	Continuous Time
Transition probabilities $P(x,y)$	Rates $\alpha(x,y)$
Matrix P	Infinitesimal generator A
$P^n(x,y) = (P^n)_{x,y}$	$P_t(x,y) = (P_t)_{x,y} = (e^{tA})_{x,y}$
Stat. dist. solves $\pi P = \pi$	Stat. dist. solves $\pi A = 0$, $\sum \pi_x = 1$

2. Conditional Expectation

Definition 2.1. Let X, Y be random variables. Assume X take values in $\mathbb{R}$ and $\mathbb{E}[|X|] < \infty$. The **conditional expectation** $\mathbb{E}[X|Y]$ is the unique random variable in $\mathbb{R}$ such that:

- $\mathbb{E}[X|Y]$ is a function of Y,
- if $F(y)$ is a function of Y in $\mathbb{R}$ with $\mathbb{E}[|F(Y)|] < \infty$, then

$$\mathbb{E}[XF(Y)] = \mathbb{E}\big[\mathbb{E}[X|Y]F(Y)\big].$$

Definition 2.2. The **conditional probability** of an event E given Y is

$$\mathbb{P}[E|Y] := \mathbb{E}[\mathbb{1}_E|Y].$$

Proposition 2.3. *$\mathbb{E}[X|\mathcal{F}_n]$ is the unique random variable in $\mathbb{R}$ such that*

- *$\mathbb{E}[X|\mathcal{F}_n]$ is $\mathcal{F}_n$-measurable,*
- *if Z is $\mathcal{F}_n$-measurable and takes values in $\mathbb{R}$, then*

$$\mathbb{E}[XZ] = \mathbb{E}\left[\mathbb{E}[X|\mathcal{F}_n]Z\right].$$

Proposition 2.4 (Properties)**.**

- *Linearity.*
- *$\mathbb{E}[X|\mathcal{F}_n] = \mathbb{E}[X]$ if X is independent from $\mathcal{F}_n$.*
- *$\mathbb{E}[X|\mathcal{F}_n] = X$ if X is $\mathcal{F}_n$-measurable.*
- *$\mathbb{E}[ZX|\mathcal{F}_n] = Z\mathbb{E}[X|\mathcal{F}_n]$ if Z is $\mathcal{F}_n$-measurable.*
- *$\mathbb{E}\left[\mathbb{E}[X|\mathcal{F}_n]\right] = \mathbb{E}[X]$.*
- *If $m \le n$, then $\mathbb{E}\left[\mathbb{E}[X|\mathcal{F}_n]|\mathcal{F}_m\right] = \mathbb{E}[X|\mathcal{F}_m]$.*
- *$|\mathbb{E}[X|\mathcal{F}_n]| \le \mathbb{E}[|X||\mathcal{F}_n]$.*

3. Discrete Time Martingales

Definition 3.1. A sequence of random variables in $\mathbb{R}$, $\{M_n\}$, is a **martingale** with respect to the filtration $\{\mathcal{F}_n\}$ if

(i) $\{M_n\}$ is $\mathcal{F}_n$-measurable for each n,
(ii) $\mathbb{E}[|M_n|] < \infty$ for each n,
(iii) For each $m \le n$, $\mathbb{E}[M_n|\mathcal{F}_m] = M_m$. Equivalently, $\mathbb{E}[M_n|\mathcal{F}_{n-1}] = M_{n-1}$.

Example 3.2.

- Suppose X_j has common mean μ. $M_n := \sum_{j=1}^n X_j - n\mu$.
- Suppose X_j are independent with mean 0 and variance σ^2. $M_n := S_n^2 - \sigma^2 n$.
- Let $\mathcal{F}_n$ be any filtration. $M_n := \mathbb{E}[X|\mathcal{F}_n]$.
- The random walk X_n on $\mathbb{Z}$. $M_n = X_n^2 - n$.

◁

3.1. The Optional Stopping Theorem.

Theorem 3.3 (Optional Stopping Theorem, Version 1)**.** *Let $\{M_n\}$ be a martingale with respect to $\{\mathcal{F}_n\}$. Let τ be a stopping time that is bounded, i.e., such that there exists a non-random $K \in \mathbb{N}_0$ such that $\mathbb{P}[\tau \le K] = 1$. Then, $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$.*

Theorem 3.4 (Optional Stopping Theorem, Version 2)**.** *Let $\{M_n\}$ be a martingale with respect to $\{\mathcal{F}_n\}$. Let τ be a stopping time such that $\mathbb{P}[\tau < \infty] = 1$, $\mathbb{E}[|M_\tau|] < \infty$, and $\lim_{n\to\infty} \mathbb{E}\left[M_n \mathbb{1}_{\{\tau > n\}}\right] = 0$. Then, $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$.*

Proposition 3.5. *If the first or second moments of M_n are uniformly bounded and $\mathbb{P}(\tau < \infty) = 1$, then*

$$\lim_{n\to\infty} \mathbb{E}[M_n \mathbb{1}_{\{\tau > n\}}] = 0.$$

Proposition 3.6. *Let $\{M_n\}$ be a martingale with respect to $\{\mathcal{F}_n\}$. Let τ be a stopping time. Then, $\{M_{n\wedge\tau}\}$ is also a martingale with respect to $\{\mathcal{F}_n\}$.*

3.2. The Martingale Convergence Theorem.

Theorem 3.7. *Let $\{M_n\}$ be a Martingale with respect to $\{\mathcal{F}_n\}$, and assume that there exists constant $C > 0$ such that $\mathbb{E}[|M_n|] \le C$ for each n. Then,*

$$\mathbb{P}\left[M_\infty := \lim_{n\to\infty} M_n \text{ exists}\right] = 1.$$

Proposition 3.8. *Other sufficient conditions for the martingale convergence theorem:*

(i) M_n is unbounded below or above uniformly in n.
(ii) The second moments of M_n are uniformly bounded in n.

Example 3.9 (Expectation of a martingale need not be preserved in the limit). Let $\{X_n\}$ be the random walk on $\mathbb{Z}$ starting at $X_0 = 0$. Let $\tau := \min\{n : X_n = 1\}$. Note that $\{X_{n\wedge\tau}\}$ is a martingale and $X_{n\wedge\tau} \le 1$. We have

$$\lim_{n\to\infty} X_{n\wedge\tau} = X_\tau = 1 \ne 0 = \mathbb{E}[X_0].$$

◁

4. Brownian Motion

Theorem 4.1. *Let $B : [0,\infty) \to \mathbb{R}$ be a random continuous function with independent, stationary increments, and such that $B_0 = 0$. Then there exists a $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ such that for any $t \ge 0$, $B_t \sim \mathcal{N}(\mu t, \sigma^2 t)$. Further, μ and σ^2 uniquely characterize the distribution of B_t.*

Definition 4.2. The **(standard) Brownian motion** is the process $\{B_t\}_{t\ge 0}$ such that $B_0 = 0$ and

(i) B_t is continuous.
(ii) For each $s < t$, $B_t - B_s \sim \mathcal{N}(0, t-s)$.
(iii) For each $s_1 \le t_1 \le \cdots \le s_k \le t_k$, the increments $B_{t_j} - B_{s_j}$ are independent.

Proposition 4.3 (Brownian Scaling)**.** *Let B be a standard Brownian motion. Let $C > 0$ be a constant. Then $\left\{C^{-1/2}B_{ct}\right\}_{t\ge 0}$ is again a standard Brownian motion.*

Proposition 4.4. *For each $t \ge 0$,*

$$\mathbb{P}\left[B \text{ is not differentiable at } t\right] = 1.$$

Note that this is weaker than B being nowhere differentiable with probability 1.

Definition 4.5. Fix $k \in \mathbb{N}$. Let $\{(X_1^n, \ldots, X_k^n)\}_{n\in\mathbb{N}}$ be a sequence of random variables in $\mathbb{R}^k$, and $(X_1, \ldots, X_k)$ be another random variable in $\mathbb{R}^k$. We say $(X_1^n, \ldots, X_k^n) \to (X_1, \ldots, X_k)$ **in distribution** if for any $a_1 < b_1, a_2 < b_2, \ldots, a_k < b_k$ such that

$$\mathbb{P}\left[X_j = a_j \text{ or } X_j = b_j\right] = 0, \quad \forall j \in \{1, \ldots, k\}$$

we have

$$\mathbb{P}\left[X_1^n \in (a_1, b_1), \ldots, X_k^n \in (a_k, b_k)\right] \longrightarrow \mathbb{P}\left[X_1 \in (a_1, b_1), \ldots, X_k \in (a_k,\right.$$

Note that $\mathbb{P}[X_j \in \{a, b\}] = 0$ is automatic is $(X_1, \ldots, X_n)$ has a continuous distribution.

It turns out that it is equivalent to requiring for each bounded and continuous $f : \mathbb{R}^k \to \mathbb{R}$ that

$$\mathbb{E}[f(X_1^n, \ldots, X_k^n)] \longrightarrow \mathbb{E}[f(X_1, \ldots, X_k)].$$

Theorem 4.6. *Let $\{X_j\}_{j\geq 0}$ be the (unbiased) random walk on $\mathbb{Z}$ with $X_0 = 0$. We extend to $X : [0,\infty) \to \mathbb{R}$ by linear interpolation. That is, define for each $t \in [j, j+1)$,*

$$X_t := (t-j)X_{j+1} + (j+1-t)X_j.$$

Then, $\{n^{-1/2}X_{nt}\}_{t>0}$ converges in distribution to the Brownian motion in the following sense: for each $t_1 \leq t_2 \leq \cdots \leq t_k$, we have

$$\left(n^{-1/2}X_{nt_1}, \ldots, n^{-1/2}X_{nt_k}\right) \longrightarrow \left(B_{t_1}, \ldots, B_{t_k}\right)$$

in distribution.

4.1. Properties.

Proposition 4.7 (Markov Property)**.** *For each $t > 0$, the process $\{B_{s+t} - B_t\}_{s\geq 0}$ is a Brownian motion independent from $\mathcal{F}_t$.*

Definition 4.8. A continuous-time stochastic process $\{M_t\}_{t\geq 0}$ in $\mathbb{R}$ is a **martingale** with respect to the filtration $\{\mathcal{F}_t\}$ if

(i) M_t is $\mathcal{F}_t$-measurable for each t.
(ii) $\mathbb{E}[|M_t|] < \infty$ for each t.
(iii) For each $s < t$, $\mathbb{E}[M_t|\mathcal{F}_s] = M_s$.

Proposition 4.9. *$\{B_t\}$ is a martingale with respect to the natural filtration $\{\mathcal{F}_t\}$.*

Proposition 4.10 (Optional Stopping Theorem)**.** *Let $\{M_t\}$ be a continuous martingale with respect to $\{\mathcal{F}_t\}$. Let τ be a stopping time and assume there exists a constant C such that*

$$\mathbb{P}\left[\tau < \infty, |M_t| \leq C, \forall t \leq \tau\right] = 1.$$

Then, $\mathbb{E}[M_t] = \mathbb{E}[M_0]$.

Proposition 4.11. *Let $a, b > 0$ be constants. Let $\tau := \min\{t \geq 0 : B_t = -a \text{ or } B_t = b\}$. Then, $\mathbb{P}[B_\tau = b] = a/(a+b)$.*

Proposition 4.12. *The Brownian motion is recurrent. That is, for each $T > 0$, we have*

$$\mathbb{P}\left[\exists t > T : B_t = x\right] = 1.$$

Proposition 4.13 (Strong Markov Property)**.** *Let τ be a stopping time. Then, $\{B_{s+\tau} - B_\tau\}_{s\geq 0}$ is a Brownian motion independent from $\mathcal{F}_\tau$.*

Proposition 4.14 (The Reflection Principle)**.** *For any $a > 0$, we have*

$$\mathbb{P}\left[\max_{0\leq s\leq t} B_s \geq a\right] = 2\mathbb{P}[B_t \geq a].$$

Equivalently, $\max_{0\leq s\leq t} B_s$ has the same distribution as $|B_t|$.

Proposition 4.15. *Let B be a one-dimensional Brownian motion. Let $\tau := \max\{t \in [0,1] : B_t = 0\}$. Note that τ is* not *a stopping time. Then,*

$$\mathbb{P}[\tau \leq t] = \frac{2}{\pi}\arcsin(\sqrt{t}).$$

4.2. Brownian Motion in Higher Dimensions.

Definition 4.16. The **(standard) Brownian motion in $\mathbb{R}^d$** is the process $B_t = (B_t^1, \ldots, B_t^d)$, where each B_t^j are iid standard Brownian motions in $\mathbb{R}$.

Proposition 4.17. *The d-dimensional Brownian motion is the unique process in $\mathbb{R}^d$ such that $B_0 = 0$ and*

(i) is continuous,
(ii) for each $s < t$, $B_t - B_s \sim \mathcal{N}(0, (t-s)I_d)$,
(iii) for each $s_1 \leq t_1 \leq \cdots \leq s_k \leq t_k$, the increments $B_{t_j} - B_{s_j}$ are independent.

Properties:

Proposition 4.18 (Brownian Scaling in $\mathbb{R}^d$)**.** *Let B be a standard Brownian motion in $\mathbb{R}^d$. Let $c > 0$ be a constant. Then $\left\{c^{-1/2}B_{ct}\right\}_{t\geq 0}$ is again a standard Brownian motion in $\mathbb{R}^d$.*

Proposition 4.19 (Markov Property)**.** *Let $\mathcal{F}_t$ be the information in $\{B_s\}_{s\subset t}$. Let τ be a stopping time for $\{\mathcal{F}_t\}$. Then $\{B_{\tau+s} - B_\tau\}_{s\geq 0}$ is a d-dimensional Brownian motion independent from τ.*

Proposition 4.20 (Rotational Invariance)**.** *Let B be a standard Brownian motion in $\mathbb{R}^d$. Let $Q \in \mathbb{R}^{d\times d}$ be an orthogonal matrix. Then, $\{QB_t\}_{t\geq 0}$ is again a standard Brownian motion in $\mathbb{R}^d$.*

Proposition 4.21 (Properties of Brownian Motion in $\mathbb{R}^2$)**.**

(i) Self-intersects in every time interval. For each $a < b$, there exists almost surely $s, t \in [a, b]$, $s \neq t$, such that $B_s = B_t$.
(ii) Neighborhood recurrence. For each $x \in \mathbb{R}^2$, $\varepsilon > 0$, and $T > 0$. There exists almost surely $t > T$ such that $|B_t - x| < \varepsilon$. Thus the range of B is dense in $\mathbb{R}^2$.
(iii) For each $x \in \mathbb{R}^2$, $\mathbb{P}(\exists t > 0, B_t = x) = 0$.

Proposition 4.22 (Properties of Brownian Motion in $\mathbb{R}^3$)**.**

(i) Self-intersection in every time interval.
(ii) Transient. $\lim_{t\to\infty}|B_t| = \infty$.

Remark 4.23. For $d = 2$, there are points in $\mathbb{R}^2$ that are hit infinitely many times. For $d = 3$, each point is hit at most twice. This is analogous to how a random walk in $\mathbb{R}^d$ is recurrent for $d = 1, 2$ and transient for $d \geq 3$. ◇

Proposition 4.24 (Properties of Brownian Motion in $\mathbb{R}^4$)**.**

(i) No self-intersection. For each $s \neq t$, $B_s \neq B_t$.
(ii) Transient.

4.3. Brownian Motion and the Heat Equation.

Definition 4.25. Let $f : \mathbb{R}^d \to \mathbb{R}$ be C^2. The **Laplacian** is

$$\Delta f := \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}.$$

Theorem 4.26. *Let $f : \mathbb{R}^d \times [0,\infty) \to \mathbb{R}$ be C^2. Assume that for each $t > 0$, there exists $c_t > 0$ such that for each $x \in \mathbb{R}^d$, $|f(x,t)| \leq c_t e^{c_t|x|}$. Let the same bound hold for the first and second spatial derivatives of f. Suppose that f solves the* ***heat equation****:*

$$\partial_t f(x,t) + \frac{1}{2}\Delta f(x,t) = 0.$$

Let B be a d-dimensional Brownian motion. Then, $M_t = f(B_t, t)$ is a martingale with respect to the natural filtration of B.

Definition 4.27. The function $f : \mathbb{R}^d \to \mathbb{R}$ is **harmonic** if $\Delta f(x) = 0$ for each $x \in \mathbb{R}^d$. A harmonic function solves the heat equation (set $f(x,t) = f(x)$ to be independent of t).

5. Applications

5.1. Random Walk on the Integer Lattice.

Proposition 5.1. *The random walk on $\mathbb{Z}^d$ is null recurrent if $d = 1, 2$ and transient if $d \geq 3$.*

Proposition 5.2. *Consider the RW on $\mathbb{Z}$. Let τ be the first time hitting $-a$ or b $(a, b > 0)$ starting from 0. Then $\mathbb{P}[X_\tau = b] = a/(a+b)$ and $\mathbb{E}[\tau] = ab$ (use $M_n = X_n^2 - n$ is a martingale).*

5.2. Biased Random Walk.

$$P(x, x+1) = p, \quad P(x, x-1) = 1-p, \quad \forall x \in \mathbb{Z}.$$

Proposition 5.3. *Let $N \geq 1$. For $x \in \{0, \ldots, N\}$, we have*

$$\mathbb{P}\left[\{X_n\} \text{ hits } N \text{ before } 0 | X_0 = x\right] = \begin{cases} \frac{\left(\frac{1-p}{p}\right)^x - 1}{\left(\frac{1-p}{p}\right)^N - 1}, & p \neq \frac{1}{2}; \\ \frac{x}{N}, & p = \frac{1}{2}. \end{cases}$$

Corollary 5.4. *For each $x \geq 1$*

$$\mathbb{P}\left[\{X_n\} \text{ hits } 0 | X_0 = x\right] = \begin{cases} 1, & p \leq \frac{1}{2}; \\ \left(\frac{1-p}{p}\right)^x, & p > \frac{1}{2}. \end{cases}.$$

Corollary 5.5. *A biased random walk is recurrent if $p = 1/2$ and transient otherwise.*

Proposition 5.6. *The symmetric random walk on $\mathbb{Z}^d$ is null recurrent if $d = 1, 2$ and transient if $d \geq 3$.*

5.3. **Queuing.**

$$P(0,1) = p, \quad P(0,0) = 1 - p,$$
$$P(x, x-1) = q(1-p), \quad P(x, x+1) = p(1-q), \quad \forall x \geq 1.$$

Recurrent if $q \geq p$.

5.4. **Galton-Watson.** Offspring distribution $\{p_k\}$ with mean μ. Extinction probability 0.

Proposition 5.7. *Let φ be the generating function for $\{p_k\}$. Then $\varphi_{X_n}(s) = \varphi^{(n)} := \varphi \circ \cdots \circ \varphi$ (n times).*

Proposition 5.8. *Assume $0 < p_0 < 1$. Then a is the smallest positive solution to $s = \varphi(s)$. Moreover, if $\mu > 1$, then $a < 1$; otherwise $a = 1$.*

5.5. **Polya's urn.** One red and one blue at time $n = 0$. Let $M_n := X_n/(n+2)$ be the fraction of red balls at time n. Then M_∞ has a Uniform$(0,1)$ distribution.

6. Generating Functions

Definition 6.1. Let Y be a random variable in $\mathbb{N}_0$. The **generating function** of Y is defined as

$$\varphi \equiv \varphi_Y : [0,\infty] \ni s \longmapsto \mathbb{E}\left[s^Y\right] = \sum_{k=0}^{\infty} \mathbb{P}[Y = k] s^k \in [0,\infty]$$

Proposition 6.2 (Properties of Generating Functions)**.**

(i) $\varphi(s) < \infty$ *for all* $s \in [0,1]$.
(ii) $\varphi(1) = \sum \mathbb{P}[Y = k] = 1$.
(iii) $\varphi(0) = \mathbb{P}[Y = 0] \cdot 0^0 = \mathbb{P}[Y = 0]$.
(iv) $\varphi'(s) = \sum_{k=0}^{\infty} k\mathbb{P}[Y = k] s^{k-1}$. *Thus* $\varphi'(1) = \mathbb{E}[Y]$.
(v) If $Y_1, \ldots, Y_m$ are independent, then

$$\varphi_{Y_1+\cdots+Y_m}(s) = \mathbb{E}\left[\prod s^{Y_j}\right] = \prod \mathbb{E}\left[s^{Y_j}\right] = \prod_{i=1}^{m} \varphi_{Y_i}(s).$$

7. Distributions

Definition 7.1. A random variable M in $\mathbb{N}$ has the **geometric distribution** with **success probability** $p \in (0,1)$ if

$$\mathbb{P}(M = m) = p(1-p)^{m-1}, \quad \forall m \geq 1.$$

Definition 7.2. A random variable T in $[0,\infty)$ has the **exponential distribution** with parameter $\lambda > 0$ if for each $t > 0$,

$$\mathbb{P}[T \geq t] = e^{-\lambda t}.$$

Proposition 7.3 (Properties of the Expoenetial Distribution)**.**

(i) $\mathbb{E}[T] = 1/\lambda$:

$$\mathbb{E}[T] = \mathbb{E}\left[\int_0^\infty \mathbb{1}(t \leq T)\,\mathrm{d}t\right] = \int_0^\infty \mathbb{E}[\mathbb{1}(t \leq T)]\,\mathrm{d}t = \int_0^\infty e^{-\lambda t}\,\mathrm{d}t = \frac{1}{\lambda}$$

(ii) Memoryless:

$$\mathbb{P}[T - t \geq s | T \geq t] = e^{-\lambda s}, \quad \forall t, s \geq 0$$

(iii) Minimum: Let $T_j \sim$ Exponential(λ_j) be independent. Then $T := \min_{1 \leq j \leq n} T_j \sim$ Exponential $\left(\sum_{j=1}^n \lambda_j\right)$ and

$$\mathbb{P}\left[T_j = \min\{T_1, \ldots, T_n\}\right] = \frac{\lambda_j}{\lambda_1 + \cdots + \lambda_n}.$$

Definition 7.4. A random variable Y in $\mathbb{N}_0$ has the **Poisson distribution** with mean $\lambda > 0$ if

$$\mathbb{P}[Y = k] = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k \in \mathbb{N}_0.$$

Lemma 7.5. *Let Y_1, Y_2 be independent Poisson random variables with means λ_1, λ_2 respectively. Then $Y_1 + Y_2 \sim$ Poisson$(\lambda_1 + \lambda_2)$.*

Definition 7.6. The **Poisson process** with rate $\lambda > 0$ is the continuous time stochastic process $\{X_t\}_{t \geq 0}$ such that $X_0 = 0$ and for each $0 \leq s_1 \leq t_1 \leq \cdots \leq s_k \leq t_k$, the random variables $X_{t_1} - X_{s_1}, \ldots, X_{t_k} - X_{s_k}$ are independent and each has distribution Poisson$(\lambda(t_j - s_j))$.

Proposition 7.7. *The inter-arrival times $\tau_j - \tau_{j-1}$ of a Poisson process are iid with distribution* Exponential(λ).

Proposition 7.8 (Stirling)**.** $n! \sim \sqrt{2\pi n} n^{n+\frac{1}{2}} e^{-n}$.

$X \sim$	supp $\mathbb{P}(X = \cdot)$	$\mathbb{P}(X = x)$	$\mathbb{E}$	$\mathbb{V}$
Binomial(n,p)	$\{0, 1, \ldots, n\}$	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Geometric(p)	$\mathbb{N}$	$(1-p)^{x-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	$\mathbb{N} \cup \{0\}$	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ

$X \sim$	supp f	f	$\mathbb{E}$	$\mathbb{V}$
Uniform(a,b)	$[a,b]$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$[0,\infty)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$\mathcal{N}(\mu, \sigma^2)$	$(-\infty, \infty)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2

Proposition 7.9 (Properties of the Normal Distribution)**.**

- *If $X \sim \mathcal{N}(0,1)$ (the **standard normal**), then $\sigma X + \mu \sim \mathcal{N}(\mu, \sigma^2)$.*
- *If $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$, and X, Y are independent, then $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.*

Definition 7.10. For $d \in \mathbb{N}$, the **Gaussian distribution** on $\mathbb{R}^d$ with mean 0 and covariance matrix $\sigma^2 I$ is the distribution $x = (x_1, \ldots, x_n) \in \mathbb{R}^d$ with density

$$\prod_{j=1}^{d} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x_j^2}{2}} = \frac{1}{(\sqrt{2\pi}\sigma)^d} e^{-\frac{|x|^2}{2\sigma^2}}.$$

Proposition 7.11. *The Gaussian distribution on $\mathbb{R}^d$ is rotational invariant.*